

PES UNIVERSITY

Department of Computer Science & Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #33

Micro-Lending & Peer Credit Risk Assessor

Project Domain: Retail, E-Commerce & Finance

Target Actors: Borrower, Peer Lender

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Prepared from the provided Lab 1 handout. The requirements below expand the supplied sample requirements into the exact submission structure requested by the handout.

1. Problem Context & System Overview

The Micro-Lending & Peer Credit Risk Assessor is a peer-to-peer micro-finance platform that calculates borrower risk scores using heuristic rules, generates structured repayment schedules, and triggers automated default warnings. The provided handout identifies **Borrower** and **Peer Lender** as the target actors.

Core idea: the platform helps a borrower present a financial profile, evaluates repayment risk, supports peer lenders in reviewing requests, creates a repayment plan, records repayments, and warns about potential defaults.

2. Functional Requirements

ID	Requirement	Priority	Description	Acceptance Criteria	Rationale
FR-001	Borrower Credit Risk Assessment	High	The system shall compute a borrower credit risk rating using debt-to-income ratio, repayment history, and employment stability indicators.	Pass: A complete profile receives a Low/Medium/High risk tier consistent with the configured heuristic rules. Fail: an incomplete financial profile is not approved for risk assessment.	Provides the core credit-risk assessment required by the problem statement and supports lender decision-making.
FR-002	Borrower Financial Profile Management	High	The system shall allow a borrower to create and update the financial information required for risk assessment, including income, existing debt, repayment history, and employment stability.	Pass: Required fields are captured and saved successfully. Fail: missing mandatory information prevents the profile from being submitted for assessment.	Ensures the risk engine receives the inputs needed to produce a meaningful risk rating.
FR-003	Peer Loan Request & Review	High	The system shall allow a borrower to submit a micro-loan request and allow a peer lender to view the request together with the borrower's risk rating and relevant repayment information.	Pass: A submitted request is visible to the intended peer lender with its risk result. Fail: an unassessed or incomplete request cannot be presented as credit-ready.	Connects the borrower's funding request with the peer lender's credit decision process.
FR-004	Repayment Schedule & EMI Processing	High	The system shall generate a structured repayment schedule for an approved loan and record repayment/EMI transactions against the schedule.	Pass: The schedule contains due dates and amounts, and each valid repayment updates the outstanding balance once. Fail: duplicate processing does not create a second repayment entry.	Supports predictable repayment and reflects the handout's requirement for structured schedules and safe transaction processing.
FR-005	Automated Default Warning	Medium	The system shall monitor scheduled repayments and trigger an automated warning when a borrower misses or becomes overdue on a required repayment.	Pass: An overdue condition produces a warning associated with the affected loan. Fail: no warning is issued when all repayments are current.	Helps borrowers and peer lenders identify repayment risk early and supports the platform's default-warning objective.

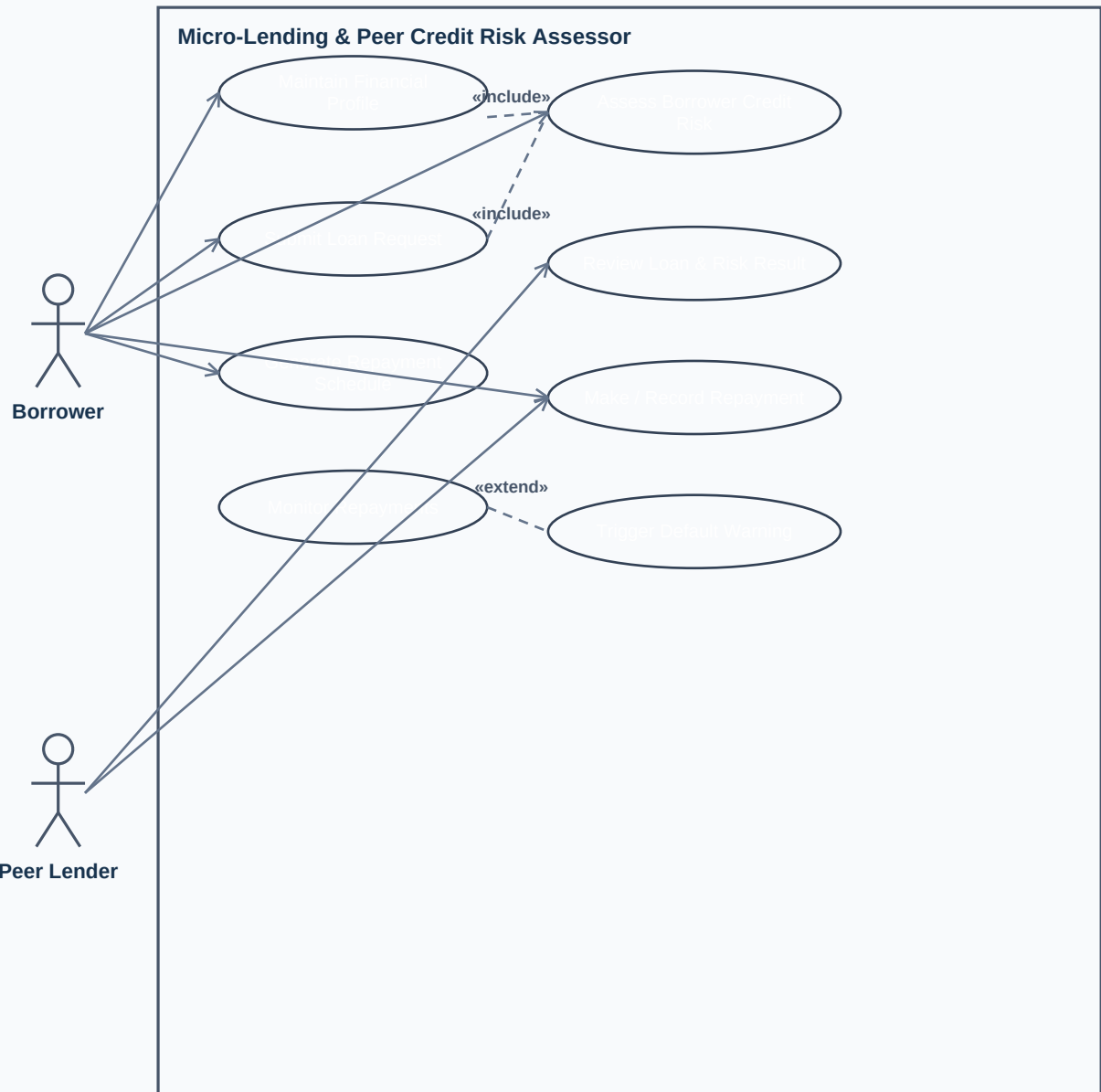
3. Non-Functional Requirements

The handout requires exactly two NFRs, each with ID, Type, Description, Priority, Acceptance Criteria, and Rationale. NFR-001 below directly follows the supplied ACID-transaction guideline; NFR-002 expands the platform's security/performance needs while staying within the problem context.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	High	Loan repayment transactions and EMI distribution shall execute inside database ACID transactions to prevent double-spending or inconsistent repayment balances.	Pass: Transaction tests show that a valid repayment is committed once and concurrent/duplicate attempts cannot create an inconsistent balance. Fail: any test produces duplicate settlement or partial updates.	The provided handout explicitly identifies ACID transactions as necessary for safe repayment and EMI distribution.
NFR-002	Security & Performance	High	The system shall protect borrower financial information from unauthorized access and return risk-assessment results within the platform's defined response-time target under simulated peak load.	Pass: Unauthorized access attempts are denied and peak-load benchmarking remains within the configured response-time target. Fail: protected borrower data is exposed or the target is exceeded.	Borrower financial data is sensitive to the operation of a credit platform, while responsive assessment is important for peer-lending decisions.

4. UML Use-Case Model

Actors: **Borrower** and **Peer Lender**. The diagram models the primary use cases needed to support the requirements. It includes the required «include» and «extend» relationships.



Actors interact with system use cases; «include» = required reused behavior, «extend» = optional/conditional behavior.

5. Use-Case Catalogue

ID	Use Case	Primary Actor(s)	Purpose
UC-01	Maintain Financial Profile	Borrower	Create/update financial information needed for credit assessment.
UC-02	Assess Borrower Credit Risk	Borrower, Peer Lender	Compute and display the borrower risk tier from the configured heuristic indicators.
UC-03	Submit Loan Request	Borrower	Submit a micro-loan request after providing required information.
UC-04	Review Loan & Risk Result	Peer Lender	Review a loan request and its risk result before making a lending decision.
UC-05	Generate Repayment Schedule	Borrower, Peer Lender	Create the structured repayment plan for an approved loan.
UC-06	Make / Record Repayment	Borrower, Peer Lender	Process a repayment/EMI and update the outstanding balance safely.
UC-07	Monitor Repayments	Borrower, Peer Lender	Track scheduled versus completed repayments.
UC-08	Trigger Default Warning	Borrower, Peer Lender	Issue a warning when an overdue repayment condition is detected.

Relationship Notes

- **Assess Borrower Credit Risk «include» Validate/maintain required financial profile information:** the risk assessment depends on required borrower inputs being available.
- **Generate Repayment Schedule «include» Calculate EMI / schedule values:** schedule creation necessarily uses the loan terms to determine repayment amounts and dates.
- **Trigger Default Warning «extend» Monitor Repayments:** the warning is conditional and occurs only when an overdue/default condition is detected.

6. Use-Case Flow Specification

Core Use Case: UC-02 — Assess Borrower Credit Risk

Primary Actor(s): Borrower; Peer Lender (consumer of the result)

Goal: Determine a Low, Medium, or High borrower risk tier using the configured heuristic indicators.

Preconditions

1. The borrower has an account/profile in the platform.
2. Required financial information is available, including income, existing debt, repayment history, and employment stability indicators.
3. The risk-assessment rules are configured and available to the system.

Main Success Scenario

1. Borrower opens the credit-risk assessment option.
2. System validates that all mandatory financial inputs are present and usable.
3. System calculates the debt-to-income ratio from the available income and debt information.
4. System evaluates repayment history and employment stability indicators.
5. System applies the configured heuristic rules to the combined indicators.
6. System assigns a risk tier: **Low, Medium, or High**.
7. System stores the assessment result with the borrower profile.
8. System makes the result available for the relevant loan request / peer-lender review.

Postconditions

- A risk tier has been assigned and stored for the borrower.
- The risk result can be associated with a loan request and viewed by the appropriate peer lender.
- No loan is treated as credit-approved solely when required financial information is incomplete.

Alternate Flow: AF-01 — Incomplete Financial Profile

1. During validation, the system detects that one or more mandatory financial indicators are missing or invalid.
2. The system stops the risk calculation.
3. The system identifies the missing information and asks the borrower to complete the profile.
4. No Low/Medium/High risk tier is issued from the incomplete profile.
5. After the borrower supplies valid information, the assessment can be initiated again.

Acceptance summary: A complete profile must produce a valid risk tier under the configured rules; an incomplete financial profile must not result in credit approval.